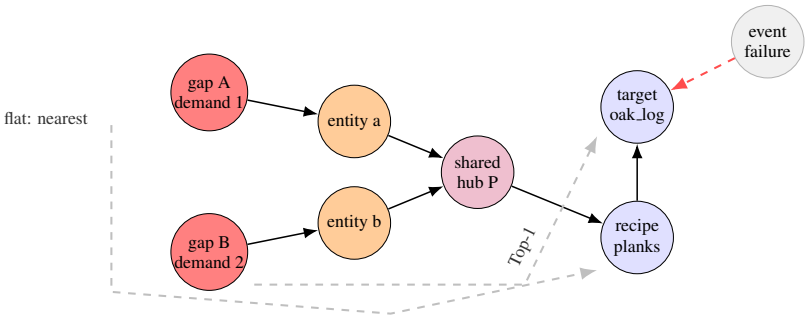


Diffusion paths: multi-source input converges via relational structure on hub P—downstream convergence (Section 4.3):
- mechanism runs: JCG>0 on 200/200 paired seeds; ≥2-input MRR@1=1.0 (hypothesis group)
- only systematic positive difference: shared-hub topology, FAS 30/30 *vs* flat 0/30 (single weight w=0.5 design)

Flat path: nearest-neighbor retrieval reaches the target directly—sufficient for ordinary retrieval:
- ΔJCG<0 under all 4 preregistered conditions (reverse-direction Holm $p=1.5\times10^{-9}$, 0/200 reversals)
- wider spread dilutes targets ($\rho=-0.445$, -0.656); *not drawn as diffusion-always-wins*.

Figure 2. Spreading activation and flat retrieval side by side. Every numbered claim comes from the archived experiments of Section 4.3; the figure only builds intuition.

Algorithm 1 Attribution–write-back: experience to relational knowledge

Require: action–observation stream (a_t, o_t) within the window; ledger $\mathscr{L}$; graph G

For each action A_t, associate observations O_k in the window with its settlement as candidate pairs (A_t, O_k)
Aggregate by action signature: $s \mathrel{+}= 1$ (positive outcome), $c \mathrel{+}= 1$ (contra)
$\text{conf} = s/(s+c+2)$ (Laplace smoothing)
if $s \geq 5$ and $\text{conf} \geq 0.80$ **then**
    create `action`:A, `change`:O nodes if absent
    write edge `action` $\longrightarrow$ `change` with relation `leads-to`, weight w=conf; mark both active
end if
ledger in parallel maintains per-action success priors $\mathscr{L}.\text{prior}(\text{sig}(A))$ consumed by action selection